

Predicting Ski Jumps using State-Space Model

Živa Hegler and Neca Camlek

JSI

May 12, 2026

(JSI - Jožef Stefan Institute)

Outline

- Introduction
- Modeling Framework and Dataset
 - State-space models and Least squares method
 - Data processing
- Methodology
 - Different approaches and error function
- Main Results
 - Animation app
- Discussion and Conclusion

Motivation

- ▶ Planning of ski jump enlargement and anticipating risky conditions
- ▶ Goal: Predict ski jumps based on external factors



Figure: Flying hill in Planica

► State-space models (SSM)

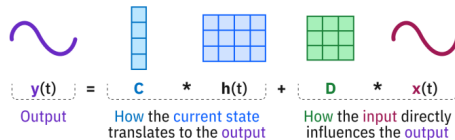
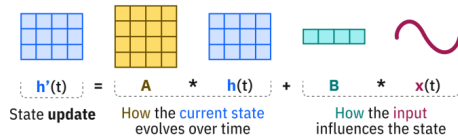
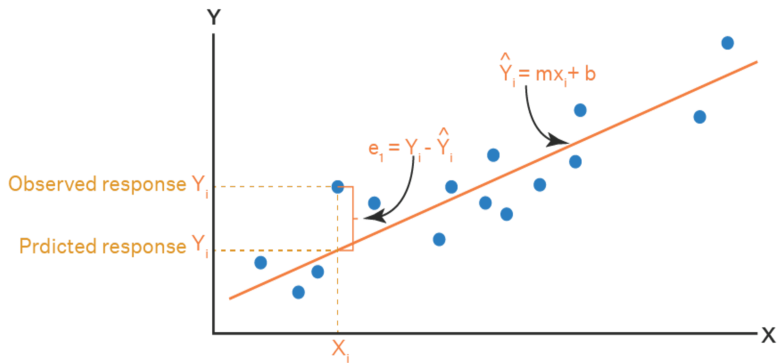


Figure: Schema of SSM

► Least squares method (LS)



Data Description

- ▶ Cleaning the data
- ▶ Data structure
 - ▶ timestamps as rows
 - ▶ 17 columns
 - ▶ Time
 - ▶ Position
 - ▶ Speed
 - ▶ Angles
 - ▶ Wind column

Data Description

- ▶ Angles: opening, stalling, yaw, roll
- ▶ Wind column WindTime|WindName|WindSpeed|Wind...

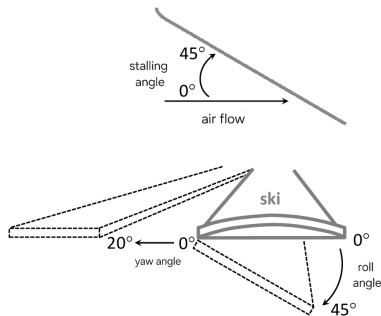
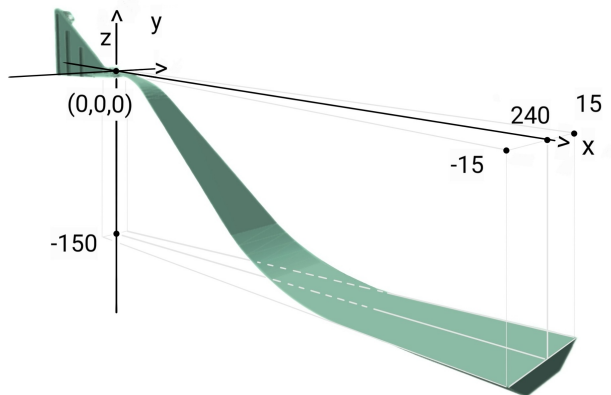


Figure: Representation of different measured angles.

Ski jumping hill

- ▶ wind features : Speed, Tangent, Cross, Turbulence
- ▶ zones
- ▶ Coordinate system on a main ski jumping hill in Planica



Different approaches

Different approaches:

- ▶ PINN
- ▶ SSM + PINN
- ▶ pure SSM:
 - ▶ State vector (x, y, z coordinates and their speed, jumpers' speed, and angles)
 - ▶ Observation vector (x, y, z , height)
 - ▶ Controls vector (averaged wind features and angles)

Model structure:

- ▶ A: state $\rightarrow$ state, B: control $\rightarrow$ state
- ▶ c: state $\rightarrow$ observation, D: control $\rightarrow$ observation

Average file and Error function

Average file starter:

- ▶ interpolation over the data points
- ▶ average number of time stamps
- ▶ first realistic input of the jump

Model's error:

- ▶ Error function
- ▶ leave-one-out cross-validation

Results

Different add-ons to the model:

- ▶ all 12 wind sensors or only 3 wind zones
- ▶ squared wind values

Model	Training Error	Test Error
SSM + wind zones	1.32 m	1.37 m
SSM + wind zones + average file	1.58 m	1.63 m
SSM	1.62 m	1.82 m
SSM + squared wind features	1.33 m	1.43 m

Actual vs Simulated Trajectory
Error: 1.361, Simulated length: 185.0, Actual length: 192.1

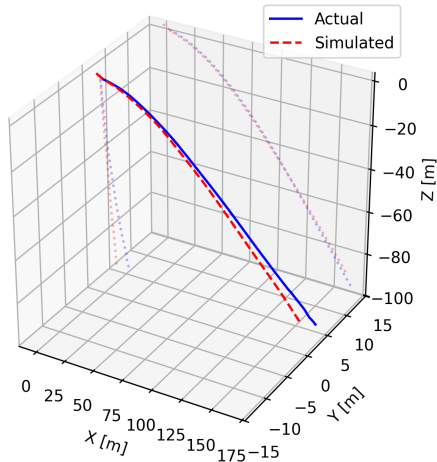


Figure: Graph with a simulated and actual flight trajectory.

Parameter analysis

A:

- ▶ X, Y, Z position -> previous state
- ▶ velocity on y axis -> yaw angles
- ▶ opening angle -> yaw angles

B:

- ▶ Landing turbulence -> angles

Application

- ▶ Description of the application
- ▶ Observing measured and predicted jumps

Ski jump animation

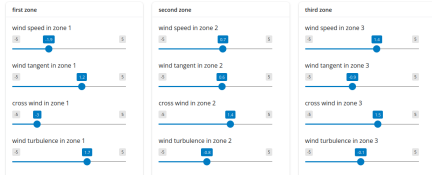
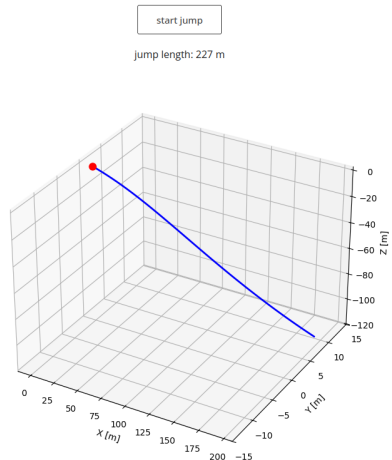


Figure: Adjusted sliders in the app



Interpretation

- ▶ Limitations
- ▶ Future work and potential improvements

Thank you!